Exercise sheet 1 – Solution

Linear Optimization (WS 26/27)

Classroom task 1.1

Consider the matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$. *How many ways do you know to characterize invertibility of* A*? Which one is the most useful to see or prove invertibility of* A*?*

Some characterizations for a square matrix $A \in \mathbb{R}^{n \times n}$ are:

1. A is invertible.
2. $\det(A) \neq 0$.
3. $\operatorname{rank}(A) = n$.
4. The columns of A are linearly independent.
5. The rows of A are linearly independent.
6. The homogeneous system $Ax = 0$ has only the solution $x = 0$.
7. For every $b \in \mathbb{R}^n$, the system $Ax = b$ has exactly one solution.

For the given matrix, the determinant test is useful. Expanding along row 1 uses the entry $a_{12} = 1$ and the minor with entries $2, 1, 1, 0$:

$$\begin{aligned} \det(A) &= (-1)^{1+2} \cdot 1 \cdot (2 \cdot 0 - 1 \cdot 1) && \text{Laplace expansion along row 1} \\ &= -1 \cdot (2 \cdot 0 - 1 \cdot 1) && (-1)^3 = -1 \\ &= -1 \cdot (0 - 1) && 2 \cdot 0 = 0 \text{ and } 1 \cdot 1 = 1 \\ &= -1 \cdot (-1) && 0 - 1 = -1 \\ &= 1 && (-1) \cdot (-1) = 1 \end{aligned}$$

Because $\det(A) = 1 \neq 0$, the matrix A is invertible.

Check with the inverse candidate

$$A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 0 \\ 2 & 1 & -2 \end{pmatrix}.$$

The product AA^{-1} has rows

$$\begin{aligned} \operatorname{row}_1(AA^{-1}) &= (1, 0, 0) && \text{row 1 of } A \text{ selects row 2 of } A^{-1} \\ \operatorname{row}_2(AA^{-1}) &= 2(-1, 0, 1) + (2, 1, -2) && \text{row 2 of } A \text{ gives } 2R_1 + R_3 \\ &= (-2, 0, 2) + (2, 1, -2) && \text{multiply row 1 by 2} \\ &= (0, 1, 0) && \text{add componentwise} \\ \operatorname{row}_3(AA^{-1}) &= (-1, 0, 1) + (1, 0, 0) && \text{row 3 of } A \text{ gives } R_1 + R_2 \\ &= (0, 0, 1) && \text{add componentwise} \end{aligned}$$

Therefore $AA^{-1} = I_3$.

The determinant criterion is the most useful one here, and A is invertible.

Exercise 1.1

Ask an AI what a linear program in standard form is. Then ask for the definitions of feasible or infeasible vector, feasible set, feasible or infeasible linear program, optimal value, bounded or unbounded linear program and optimal solution. Compare the answers to the lecture notes.

(1)

AI answer, key part:

$$\begin{aligned} &\text{maximize} && c^T x \\ &\text{subject to} && Ax = b, \\ & && x \geq 0, \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$, and $x \in \mathbb{R}^n$ is the decision variable.

Lecture notes, Definition 1.11:

$$\begin{aligned} &\text{maximize} && c^T x \\ &\text{subject to} && Ax = b, \\ & && x \geq 0. \end{aligned}$$

Comparison:

Point	AI answer	Lecture notes
Objective	maximize $c^T x$	maximize $c^T x$
Constraints	$Ax = b$	$Ax = b$
Sign condition	$x \geq 0$	$x \geq 0$

The AI answer matches the lecture definition of standard form.

(2)

For a linear program in standard form, let

$$C = \{x \in \mathbb{R}^n \mid Ax = b,\ x \geq 0\}.$$

Term	AI answer, key part	Comparison with lecture notes
feasible vector	x is feasible if it satisfies all constraints $Ax = b$ and $x \geq 0$; otherwise it is infeasible.	Same as Definition 1.15.
feasible set	The set of all feasible vectors is $C = \{x \in \mathbb{R}^n \mid Ax = b,\ x \geq 0\}$.	Same as Definition 1.15.
feasible LP	The LP is feasible if $C \neq \emptyset$; otherwise it is infeasible.	Same as Definition 1.15.
optimal value	$v = \sup\{c^T x \mid x \in C\}$, with $v = -\infty$ when $C = \emptyset$.	Same as Definition 1.15.
unbounded LP	A feasible LP is unbounded if for every $M \in \mathbb{R}$ there is an $x \in C$ with $c^T x > M$; then $v = +\infty$.	Same as Definition 1.15.
bounded LP	A feasible LP is bounded if there is an $M \in \mathbb{R}$ with $c^T x \leq M$ for every $x \in C$.	Same as Definition 1.15.
optimal solution	A feasible point $x^* \in C$ is an optimal solution if $c^T x^* \geq c^T x$ for every $x \in C$.	Same as Definition 1.15.

The AI definitions match the lecture definitions when the lecture standard form is used.

Exercise 1.2

A student has at most 30 *minutes to consult sources about the course requirements. The sources have values and times: colleague* $(4, 5)$*, first lecture* $(8, 9)$*, Moodle description* $(7, 7)$*, LV-Karte* $(16, 15)$*, doing nothing* $(40, 30)$*. At least two sources must be used. Maximize personal value within the time limit. Formulate the optimization problem.*

Define the decision variables

$$x_i = \begin{cases} 1, & \text{if source } i \text{ is used,} \\ 0, & \text{if source } i \text{ is not used,} \end{cases} \qquad i = 1, \ldots, 5.$$

The values give the objective function:

$$\text{value} = 4x_1 + 8x_2 + 7x_3 + 16x_4 + 40x_5 \qquad \text{use the given values}$$

The time limit gives the time constraint:

$$5x_1 + 9x_2 + 7x_3 + 15x_4 + 30x_5 \leq 30 \qquad \text{use the given times and the 30 minute limit}$$

The requirement to use at least two sources gives

$$x_1 + x_2 + x_3 + x_4 + x_5 \geq 2. \qquad \text{count the chosen sources}$$

The binary restrictions are

$$x_i \in \{0, 1\}, \qquad \text{each source is either chosen or not chosen, } i = 1, \ldots, 5$$

Thus the optimization problem is

$$\boxed{\begin{array}{ll} \text{maximize} & 4x_1 + 8x_2 + 7x_3 + 16x_4 + 40x_5 \\ \text{subject to} & 5x_1 + 9x_2 + 7x_3 + 15x_4 + 30x_5 \leq 30, \\ & x_1 + x_2 + x_3 + x_4 + x_5 \geq 2, \\ & x_i \in \{0,1\}, \quad i = 1, \ldots, 5. \end{array}}$$

Consistency check:

$$\begin{aligned} 5 \cdot 1 + 9 \cdot 1 + 7 \cdot 1 + 15 \cdot 0 + 30 \cdot 0 = 21 \quad & \text{test sources } 1, 2, 3 \\ 21 \leq 30 \quad & \text{time limit is satisfied} \\ 1 + 1 + 1 + 0 + 0 = 3 \quad & \text{count selected sources} \\ 3 \geq 2 \quad & \text{at least two sources are used} \end{aligned}$$

The model above is the required binary optimization problem.

Exercise 1.3

FreshShine produces shower gel, shampoo and body lotion. Profits per unit are 3, 5 *and* 4 *euros. Machine times per unit are: shower gel uses* 1 *hour on A and* 1 *hour on B; shampoo uses* 2 *hours on A,* 1 *hour on B and* 3 *hours on C; body lotion uses* 1 *hour on A,* 2 *hours on B and* 1 *hour on C. Monthly capacities are A:* 170*, B:* 150*, C:* 180 *hours. All produced units can be sold. Maximize profit and formulate the mathematical optimization model.*

Define

$x_1 =$ units of Citrus Shower Gel, $x_2 =$ units of Citrus Shampoo, $x_3 =$ units of Citrus Body Lotion.

The profit is

$$\text{profit} = 3x_1 + 5x_2 + 4x_3 \qquad \text{profit per unit times number of units}$$

The capacity constraints are obtained machine by machine:

$$\begin{aligned} x_1 + 2x_2 + x_3 \leq 170 \quad & \text{machine A hours} \\ x_1 + x_2 + 2x_3 \leq 150 \quad & \text{machine B hours} \\ 0x_1 + 3x_2 + x_3 \leq 180 \quad & \text{machine C hours} \end{aligned}$$

The production quantities cannot be negative:

$$x_1, x_2, x_3 \geq 0. \qquad \text{units produced}$$

Therefore the optimization model is

$$\boxed{\begin{array}{ll} \text{maximize} & 3x_1 + 5x_2 + 4x_3 \\ \text{subject to} & x_1 + 2x_2 + x_3 \leq 170, \\ & x_1 + x_2 + 2x_3 \leq 150, \\ & 3x_2 + x_3 \leq 180, \\ & x_1, x_2, x_3 \geq 0. \end{array}}$$

Consistency check with the sample plan $(x_1, x_2, x_3) = (10, 20, 30)$:

$$\begin{aligned} 10 + 2 \cdot 20 + 30 &= 80 && \text{machine A use} \\ 80 &\leq 170 && \text{A capacity} \\ 10 + 20 + 2 \cdot 30 &= 90 && \text{machine B use} \\ 90 &\leq 150 && \text{B capacity} \\ 3 \cdot 20 + 30 &= 90 && \text{machine C use} \\ 90 &\leq 180 && \text{C capacity} \end{aligned}$$

The model above is the required linear optimization model.